

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$\begin{aligned} a. \quad \sum_{t=1}^T (t - \bar{t})^2 &= \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} \\ &= T(T+1) \left[\frac{4T+2}{12} - \frac{3T+3}{12} \right] \\ &= \frac{1}{12} T(T+1)(T-1) \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{var}(b_2|\mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (t - \bar{t})^2} \\ &= \frac{12\sigma^2}{T(T+1)(T-1)} \end{aligned}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T+1)(T-1)} = 0$$

$$\begin{aligned} b. \quad \sum_{t=1}^T (0.5^t - \overline{0.5^t})^2 &= \sum_{t=1}^T \left(\frac{1}{4}\right)^t - \left(\sum_{t=1}^T \left(\frac{1}{2}\right)^t\right)^2 / T \\ &= \frac{\frac{1}{4}(1 - (\frac{1}{4})^T)}{1 - \frac{1}{4}} - \left(\frac{\frac{1}{2}(1 - (\frac{1}{2})^T)}{1 - \frac{1}{2}}\right)^2 / T \\ &= \frac{1 - \frac{1}{4^T}}{3} - \left(1 - \frac{1}{2^T}\right)^2 / T \\ &= \frac{1}{3} - \frac{1}{3 \cdot 4^T} - \frac{1}{T} + \frac{1}{T \cdot 2^{T-1}} + \frac{1}{T \cdot 2^{2T}} \end{aligned}$$

So

$$\begin{aligned} \text{var}(b_2|\mathbf{x}) &= \frac{\sigma^2}{\sum_{t=1}^T (0.5^t - \overline{0.5^t})^2} \\ &= \frac{\sigma^2}{\frac{1}{3} - \frac{1}{3 \cdot 4^T} - \frac{1}{T} + \frac{1}{T \cdot 2^{T-1}} + \frac{1}{T \cdot 2^{2T}}} \end{aligned}$$

and

$$\lim_{T \rightarrow \infty} \text{var}(b_2 | X) = \frac{\sigma^2}{(\frac{1}{3})} = 3\sigma^2$$

C. The above results show that if our sample X is an arithmetic progression, then the variance of the estimator b_2 approaches to 0 as the number of the sample approaches to ∞ .

Meaning that the length of the confidence interval can be arbitrary close to 0 as long as the number of the sample is sufficiently large, which leads to a better prediction.

On the other hand, if our sample X is $(0.5, 0.5^2, \dots, 0.5^T)$, then the variance of the estimator b_2 approaches to $3\sigma^2$ as T approaches to ∞ , which means that even if the number of samples approaches to ∞ , the length of the confidence interval doesn't approach to 0, unlike the first case.

Thus, it shows that increasing the number of the samples doesn't necessary guarantee that $\text{var}(b_2 | X)$ approaches to 0, the formulation of the sample also matters.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a. $\beta_2 = 0$
- b. $\beta_1 + 2\beta_2 = 5$
- c. $\beta_1 - \beta_2 + \beta_3 = 4$

5.6

a. ① $H_0: \beta_2 = 0, H_a: \beta_2 \neq 0$

② $\alpha = 0.05$

③ Test statistics: $\frac{b_2 - \beta_2}{\text{se}(b_2)} \sim t_{N-3}$

④ Realized statistics: $t^* = \frac{3}{\sqrt{4}} = 1.5$

⑤ Rejection region: $\{t^* < -2 \text{ or } t^* > 2\}$ ($t_{60, 0.025} \approx 2.000$)

⑥ Since 1.5 doesn't in the rejection region, we can't reject the null hypothesis.

b. ① $H_0: \beta_1 + 2\beta_2 = 5, H_a: \beta_1 + 2\beta_2 \neq 5$

② $\alpha = 0.05$

③ Test statistics: $\frac{b_1 + 2b_2 - (\beta_1 + 2\beta_2)}{\text{se}(b_1 + 2b_2)} \sim t_{N-3}$

④ $\widehat{\text{var}}(b_1 + 2b_2) = [1 \ 2 \ 0] \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = [-1 \ 6 \ 1] \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = 11$

Realized statistics: $t^* = \frac{3}{\sqrt{11}} \approx 0.905$

⑤ Rejection region: $\{t^* < -2 \text{ or } t^* > 2\}$ ($t_{60, 0.025} \approx 2.000$)

⑥ Since 0.905 doesn't in the rejection region, we can't reject the null hypothesis.

C. ① $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, $H_a: \beta_1 - \beta_2 + \beta_3 \neq 4$,

② $\alpha = 0.05$

③ Test statistics: $\frac{(b_1 - b_2 + b_3) - (\beta_1 - \beta_2 + \beta_3)}{se(b_1 - b_2 + b_3)} \sim t_{N-3}$

④ $\hat{var}(b_1 - b_2 + b_3) = [1 \ -1 \ 1] \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = [6 \ -6 \ 4] \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 16$

Realized statistics: $t^* = \frac{-6}{\sqrt{16}} \approx -1.5$

⑤ Rejection region: $\{t^* < -2 \text{ or } t^* > 2\}$ ($t_{60, 0.025} \approx 2.000$)

⑥ Since -1.5 doesn't in the rejection region, we can't reject the null hypothesis.

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2, mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2, mean}$ is unbiased.

a. For $j=1, 2$, we have $\bar{y}_j = \beta_1 + \beta_2 \bar{x}_j + \bar{e}_j$, where $\bar{e}_j$ is defined as $\bar{y}_j - (\beta_1 + \beta_2 \bar{x}_j)$.

Then we have $(\bar{y}_2 - \bar{y}_1) = \beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$

Thus $\hat{\beta}_{2, mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$

$$\begin{aligned} \Rightarrow E[\hat{\beta}_{2, mean} | X] &= \beta_2 + \frac{E[\bar{e}_2 | X] - E[\bar{e}_1 | X]}{\bar{x}_2 - \bar{x}_1} \\ &= \beta_2 + \frac{0 - 0}{\bar{x}_2 - \bar{x}_1} \quad (\text{by MR2}) \\ &= \beta_2 \end{aligned}$$

So $\hat{\beta}_{2, mean}$ is unbiased.

b. Derive an expression for $\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{x})$.

c. Write down an expression for $\text{var}(\hat{\beta}_{2,\text{mean}})$.

b. Since $\hat{\beta}_{2,\text{mean}} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{X}_2 - \bar{X}_1}$,

we have $\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{X}) = \frac{\text{var}(\bar{e}_2 - \bar{e}_1|\mathbf{X})}{(\bar{X}_2 - \bar{X}_1)^2}$

By MR3 and MR4, we have

$$\text{var}(\bar{e}_j|\mathbf{X}) = \frac{1}{N/2} \sigma^2 = \frac{2\sigma^2}{N} \quad \text{for } j=1,2$$

and thus $\text{var}(\bar{e}_1 - \bar{e}_2|\mathbf{X}) = \frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} + 0 = \frac{4\sigma^2}{N}$.

Therefore $\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{X}) = \frac{4\sigma^2}{N(\bar{X}_2 - \bar{X}_1)^2}$

c. $\text{var}(\hat{\beta}_{2,\text{mean}}) = E[\text{var}(\hat{\beta}_{2,\text{mean}}|\mathbf{X})]$

$$= E\left[\frac{4\sigma^2}{N(\bar{X}_2 - \bar{X}_1)^2}\right]$$

$$= \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{X}_2 - \bar{X}_1)^2}\right]$$

- d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?

d. We want to find conditions such that

$$\text{var}(\hat{\beta}_{2,mean} | \mathbf{x}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \text{ approaches to } 0$$

So we need the condition $N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$

For example, if x comes from a uniform i.i.d, then $\hat{\beta}_{2,mean}$ will be a consistent estimator for β_2 .

- e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

- $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
- Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

```
> set.seed(123)
>
> sigma2 <- 1000
> sample_sizes <- c(100,500,1000,5000)
>
> results <- list()
>
> for (N in sample_sizes) {
+   x <- runif(N, min = 0, max = 10)
+   x_sorted <- sort(x)
+
+   x1 <- x_sorted[1:N/2]
+   x2 <- x_sorted[(N/2 + 1):N]
+
+   xbar1 <- mean(x1)
+   xbar2 <- mean(x2)
+
+   sx2 <- mean((x - mean(x))^2)
+
+   var_ols <- sigma2 / (N * sx2)
+   diff <- xbar2 - xbar1
+   var_mean <- 4 * sigma2 / (N * diff^2)
+
+   inv_sx2 <- 1 / sx2
+   inv_x2x1 <- 4 / (diff^2)
+
+   cat("\nSample size N =", N, "\n")
+   cat("var(b2|X)          =", var_ols, "\n")
+   cat("var(beta_2mean|X)   =", var_mean, "\n")
+   cat("E[(sx^2)^(-1)]       =", inv_sx2, "\n")
+   cat("E[4/(xbar2-xbar1)^2] =", inv_x2x1, "\n")
+ }
```

Sample size N = 100	
var(b2 X)	= 1.243625
var(beta_2mean X)	= 1.648725
E[(sx^2)^(-1)]	= 0.1243625
E[4/(xbar2-xbar1)^2]	= 0.1648725
Sample size N = 500	
var(b2 X)	= 0.244024
var(beta_2mean X)	= 0.3254221
E[(sx^2)^(-1)]	= 0.122012
E[4/(xbar2-xbar1)^2]	= 0.1627111
Sample size N = 1000	
var(b2 X)	= 0.1229063
var(beta_2mean X)	= 0.1654635
E[(sx^2)^(-1)]	= 0.1229063
E[4/(xbar2-xbar1)^2]	= 0.1654635
Sample size N = 5000	
var(b2 X)	= 0.02413174
var(beta_2mean X)	= 0.03230971
E[(sx^2)^(-1)]	= 0.1206587
E[4/(xbar2-xbar1)^2]	= 0.1615486

- f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- g. Show that $E[(S_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]
- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

f. No matter what sample size it is, we can see that $\text{var}(b_2|X) < \text{var}(\hat{\beta}_{2,mean}|X)$ and $E[(S_x^2)^{-1}] < E[4/(\bar{x}_2 - \bar{x}_1)^2]$.

As sample size increases, both $\text{var}(b_2|X)$ and $\text{var}(\hat{\beta}_{2,mean}|X)$ decreases, while $E[(S_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ remain steady.

So it appears that $\hat{\beta}_{2,mean}$ is consistent.

g. Since $\text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$, we have

$$E[(S_x^2)^{-1}] \xrightarrow{p} E\left[\left(\frac{100}{12}\right)^{-1}\right] = 0.12$$

Since $\bar{X}_1, \bar{X}_2$ are the sample mean of the 1st and the 2nd halves of a uniform distribution on the interval $(0, 10)$, we have $\bar{X}_1 \rightarrow \frac{5-0}{2} = 2.5$ and $\bar{X}_2 \rightarrow \frac{10-5}{2} = 7.5$

$$\text{So } E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow E\left[\frac{4}{25}\right] = 0.16$$

h. The denominator of $\text{var}(\hat{\beta}_{2,mean}|X)$ is $N(\bar{x}_2 - \bar{x}_1)^2$

If the variance of the data X is σ^2 , then the variance of $\bar{X}_j$ is $\left(\frac{N}{2}\right)^{-1}\sigma^2 = \frac{2\sigma^2}{N}$ for $j=1, 2$.

So $\bar{X}_j \sim O_p\left(\frac{1}{\sqrt{N}}\right)$ for $j=1, 2$, where O_p denotes the big O notation in probability. Thus $\bar{X}_2 - \bar{X}_1 \sim O_p\left(\frac{1}{\sqrt{N}}\right)$ and thus

$$N(\bar{x}_2 - \bar{x}_1)^2 \sim O_p(1),$$

which means that $\text{var}(\hat{\beta}_{2,mean}|X)$ doesn't approaches to 0 as N approaches to ∞ . In other words, the estimator $\hat{\beta}_{2,mean}$ isn't consistent in this case.